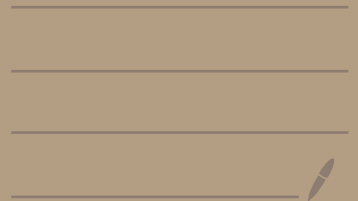


Lec 3. • Second order linear eqn.

- Homogeneous eqn with constant coe.
- Reduction of order



(I) General form

$$F[t, y, y'(t), y''(t)] = 0$$

• Rewrite it as

$$\frac{d^2 y}{dt^2} = f[t, y, y'(t)] \quad (1)$$

• Consider (simplest) case: f is linear

$$f[t, y, y'] = -p(t)y'(t) - q(t)y + g(t).$$

Then (1) $\Leftrightarrow$

$$y'' + p y' + q y = g(t) \quad (2)$$

• Or can consider:

$$P(t)y'' + Q(t)y' + R(t)y = G(t) \quad (2')$$

with $P(t) \neq 0$

$$(2') \Leftrightarrow (2) \quad v := a \quad p = \frac{Q}{P}, \quad q = \frac{R}{P}, \quad g = \frac{G}{P}.$$

If f is not in the form (2), (2'), then (1) is nonlinear. In general, no explicit formula.

(II) Basic concept.

• Initial Value: E.g. $y'' = 0$

$$\Rightarrow y = C_1 t + C_2. \quad [\text{General Soln}]$$

Need 2 conditions:

$$y(t_0) = y_0, \quad y'(t_0) = y_0'. \quad (3)$$

(2) + (3) call Initial value problem.

• Homogeneous: If $f(t) \equiv 0$ in ②.

In this case, if $y(t)$ solves ②,

for any constant λ , $\lambda y(t)$ also solves ②

• Nonhomogeneous, if $f(t) \neq 0$

(III): Homogeneous eqn with constant coefficient.

consider
 $f \equiv 0$

$P(t) \equiv a$, $Q(t) \equiv b$, $R(t) \equiv c$, a, b, c constant.

$$ay'' + by' + cy = 0 \quad (4)$$

• Ansatz: want y'', y', y have similar form.

$$y = e^{\lambda t}, \quad y' = \lambda e^{\lambda t}, \quad y'' = \lambda^2 e^{\lambda t}.$$

$e^{\lambda t}$ solves ④ $\Leftrightarrow$ characteristic eqn:

$$[a\lambda^2 + b\lambda + c] e^{\lambda t} = 0$$

$$\Rightarrow a\lambda^2 + b\lambda + c = 0 \quad (5)$$

In general, there are 2 solutions λ_1, λ_2 .

Then both

$$y_1(t) = e^{\lambda_1 t}, \quad y_2(t) = e^{\lambda_2 t} \text{ solves } (4)$$

General solution ($\lambda_1 \neq \lambda_2$)

$$y(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

To determine C_1, C_2 , needs 2 more conds.

e.g. $y(t_0) = y_0, y'(t_0) = y'_0$.

E.g. [do examples]

For $a, b, c \in \mathbb{R}$, in general, we have 3 cases

- (1) $\lambda_1 \neq \lambda_2$ real ✓ $y(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$
- (2) λ_1, λ_2 complex. ?? if we want real solution.
- (3) $\lambda_1 = \lambda_2$, repeated root. (they must be real ?? why ??).
only 1 solution.

(IV). Complex root.

λ_1, λ_2 .

$$y(t) = \underbrace{C_1 e^{\lambda_1 t}}_{\text{complex}} + \underbrace{C_2 e^{\lambda_2 t}}_{\text{complex}} \quad (5)$$

want to rewrite (5) as real.

$$\lambda_1 = \lambda + i\mu, \quad \lambda_2 = \bar{\lambda}_1 = \lambda - i\mu.$$

Euler formula: $\forall x \in \mathbb{R}$

$$\begin{cases} e^{ix} = \cos x + i \sin x \\ e^{-ix} = \cos x - i \sin x. \end{cases}$$

Thus.

$$e^{\lambda_1 t} = e^{(\lambda + i\mu)t} = e^{\lambda t} [\cos \mu t + i \sin \mu t]$$

$$e^{\lambda_2 t} = e^{(\lambda - i\mu)t} = e^{\lambda t} [\cos \mu t - i \sin \mu t]$$

$$\Rightarrow y(t) = (C_1 + C_2) e^{\lambda t} \cos \mu t + (C_1 - C_2)i e^{\lambda t} \sin \mu t.$$

upon changing $C_1, C_2 \Rightarrow$ general solution in real value

$$y(t) = \underbrace{\tilde{C}_1 e^{\lambda t} \cos(\mu t)}_{y_1(t)} + \underbrace{\tilde{C}_2 e^{\lambda t} \sin(\mu t)}_{y_2(t)}$$

Ex: If $\lambda \pm i\mu$ solves

$$ax^2 + bx + c = 0$$

verify

$y_1(t), y_2(t)$ solves

$$ay'' + by' + c = 0.$$

Ex: $y'' + y = 0 \quad \lambda = \pm i$

$$\Rightarrow y_1(t) = \cos t, \quad y_2(t) = \sin t$$

(V) Repeated Root. $\lambda_1 = \lambda_2$.

In this case, $y_1(t) = y_2(t) = e^{\lambda_1 t}$
but expect 2 different solutions.

(V.1) Reduction of order:

Consider general 2nd order linear ODE.

$$P(t) y''(t) + Q(t) y'(t) + R(t) y = 0. \quad (6)$$

P, Q, R not necessarily constants.

• Suppose $y_1(t)$ solves (6)

• Find 2nd solution by

$$y_2(t) = \mu(t) y_1(t), \quad \mu(t) \neq 0 \text{ unknown.}$$

• y_2 s (6) $\Rightarrow$

$$P(\mu y_1)'' + Q(\mu y_1)' + R \mu y_1 = 0$$

$$\Leftrightarrow P[\mu'' y_1 + 2\mu' y_1' + \underline{\mu y_1''}] + Q[\mu' y_1 + \underline{\mu y_1'}] + R \mu y_1 = 0$$

$$\Leftrightarrow P \mu'' y_1 + [2P y_1' + Q y_1] \mu' + \underbrace{[P y_1'' + Q y_1' + R y_1]}_{=0 \text{ since } y_1 \text{ solves (6)}} \mu = 0$$

$$\Leftrightarrow P \mu'' y_1 + [2P y_1' + Q y_1] \mu' = 0$$

(7)

(6)

• 1st order linear ODE on μ'

$$\mu' = \exp \left[- \int \frac{2P y_1' + Q y_1}{P y_1} ds \right]$$

• Further integrate $\Rightarrow \mu(t) \Rightarrow y_2(t) = \mu(t) y_1(t)$.

• Apply to $ay_1'' + by_1' + cy_1 = 0$, $P \equiv a$, $Q \equiv b$, $R \equiv c$

$$\textcircled{7} \Rightarrow a\mu'' \cdot e^{\lambda_1 t} + (2a(e^{\lambda_1 t})' + b e^{\lambda_1 t})\mu' = 0$$

$$\Leftrightarrow a\mu'' e^{\lambda_1 t} + (2a\lambda_1 + b)e^{\lambda_1 t}\mu' = 0$$

$$\text{Since } \lambda_1 = \lambda_2 = -\frac{b}{2a} \Rightarrow$$

$$a\mu'' = 0 \Rightarrow \mu'' = 0$$

Take $\mu(t) = t$, get

$$y_2(t) = t \cdot y_1(t) = t e^{\lambda_1 t}$$

General solution

$$y(t) = C_1 e^{\lambda_1 t} + C_2 t e^{\lambda_1 t}$$

$$\lambda_1 = -\frac{b}{2a} \in \mathbb{R}$$

Ex.:

$$2t^2 y'' + 3t y' - y = 0, \quad y_1(t) = t^{-1}$$

Find $y_2(t)$.